Alex Vedernikov

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Hiring Team, Data Science
Doktar

Re: Senior Geospatial Data Scientist (Istanbul, hybrid)

Dear Hiring Team,

Geospatial data is what I most want to work on, and a role that turns spatial and temporal signals into decisions for farmers, corporations and governments is a good fit for that. I am a Senior Data Scientist with a PhD in Computer Science (optimal route planning) and 11 years of experience, much of it on geospatial data, and relocation is possible. I hold Australian and Ukrainian citizenship.

At GapMaps, a Melbourne location-intelligence company, I was a Senior Data Scientist working on analytics for clients across fast food, fitness, childcare and retail. I set up a pipeline ingesting around 100M device locations per day and extracted spatio-temporal visitation patterns for retail areas and shopping malls in nine countries. I also developed a fully automated pipeline producing time-sensitive street-traffic metrics, which replaced expensive third-party solutions, and built a country-specific retail hierarchy of shopping malls from multi-source data. On the imagery side, I conducted an analysis of swimming-pool detection and incorporated the results into our framework for identifying affluent areas, estimating consumer classes in combination with demographic data. I was responsible for purchasing mobile location data, so I know how to assess data bias and fit for a given need. The stack was Python, Snowflake, AWS, PostGIS, QGIS and ArcGIS.

At Aurecon, a large engineering consultancy, I was a Data Scientist on geospatial projects. I used deep learning to detect road kerbs on satellite imagery and compared the detections with cadastral data, integrating both sources to update cadastral GIS data for Wellington City Council. In addition, I used around 500M user mobile locations to find visitation patterns at Sydney beaches for Sydney Water, conducted a spatial analysis of customer behaviour for a government transport authority, and built the data pipeline for a port scheduling optimisation project, using GeoPandas, Apache Sedona and Databricks. The consulting work taught me to turn analysis into recommendations for non-technical stakeholders. I also supervised two interns.

At DayTrip, a travel marketplace, I was a Senior Data Scientist and built an automated pricing system that added around 600K EUR of revenue per year relative to standard prices. I unified the data across spatio-temporal dimensions such as country, route and time to departure, and designed the data layers, from raw data to dashboard-ready, that several teams used. The pricing work also had a geospatial component: I analysed driver distances with the Google Maps API as part of load optimisation.

To be upfront about gaps: my geospatial work has centred on location and mobility data, with satellite imagery used in two projects (kerb detection and affluent-area detection) and hands-on experience with Sentinel satellite imagery. I have not worked hands-on with climate and Earth observation datasets such as MODIS, GRACE, ERA5 or CHIRPS, with Google Earth Engine or Xarray, or on hydrology and water-balance problems. I am strong in Python, geospatial data pipelines, spatio-temporal modelling and communicating results. I adapt quickly to new domains, having moved from transport consulting to retail location intelligence to travel pricing, and I keep studying: I have completed more than ten courses, including recent ones on agentic AI.

I look forward to discussing how I can contribute to Doktar's work.

Kind regards,
Alex Vedernikov